

OPTICALLY STIMULATED LUMINESCENCE OF EUROPIUM DOPED LAP

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1. Overview

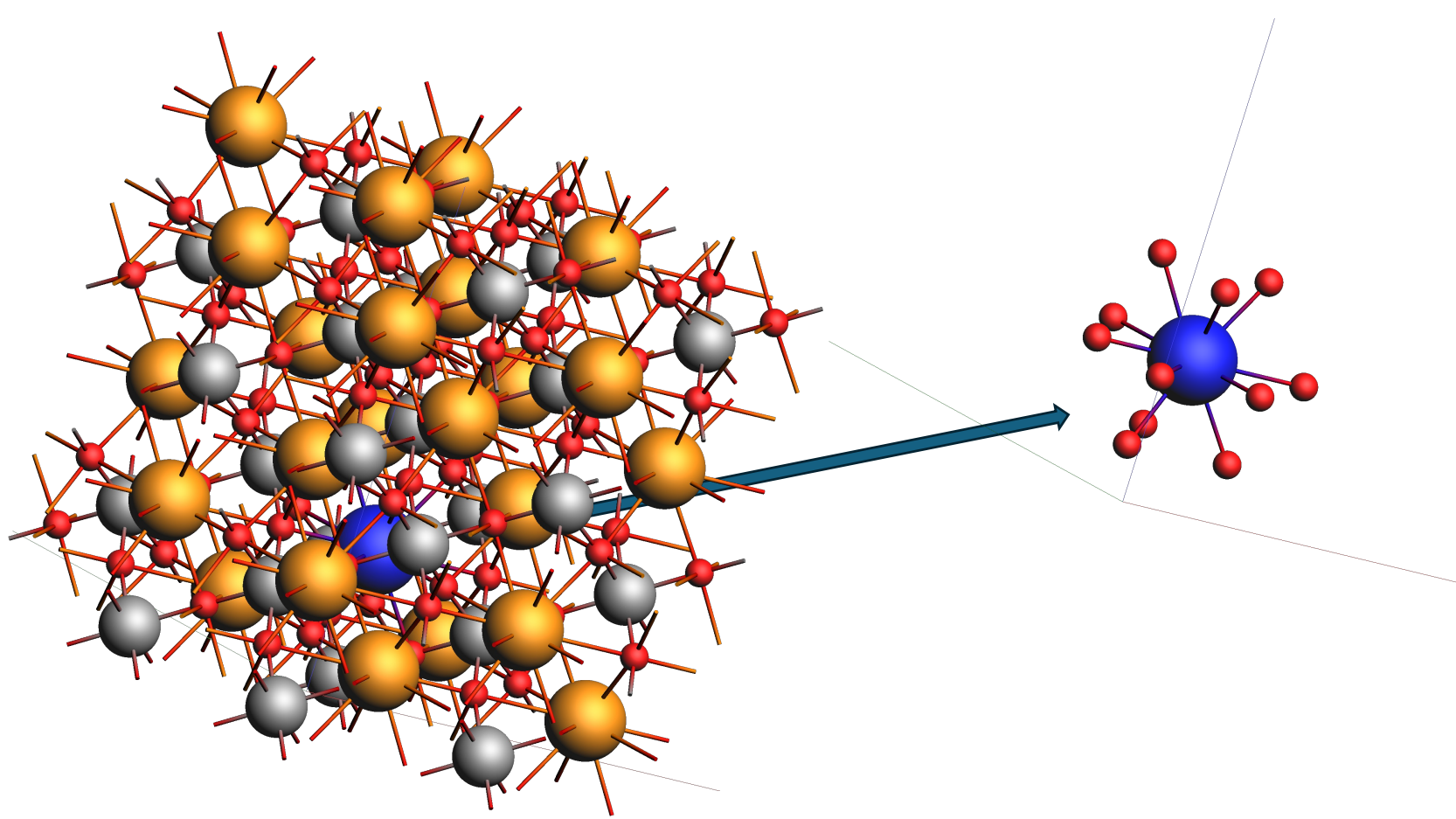
LaAlO₃(LAP) doped with Europium is a phosphor widely used in lighting (LED) and display technologies and medical imaging, such as CT(computed tomography) and X-ray imaging.

In this work, we did the following:

- Compared the calculation of energy levels of Eu with the measured emission spectrum.
- Measured the decay dynamics of the excited Eu population after UV excitation.
- Evaluated the fit of a simple model.

2. Sample

1. Crystal Structure of LaAlO₃:Eu³⁺



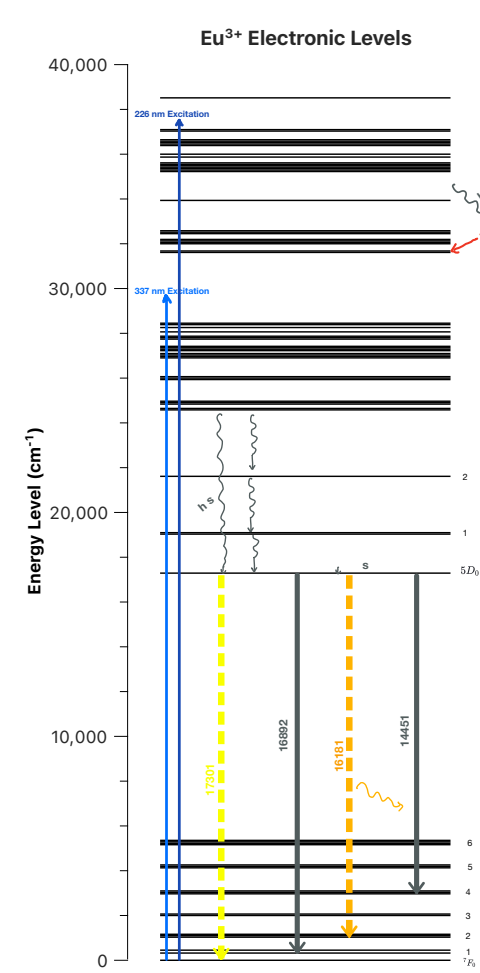
2. Mounted Sample



Room light

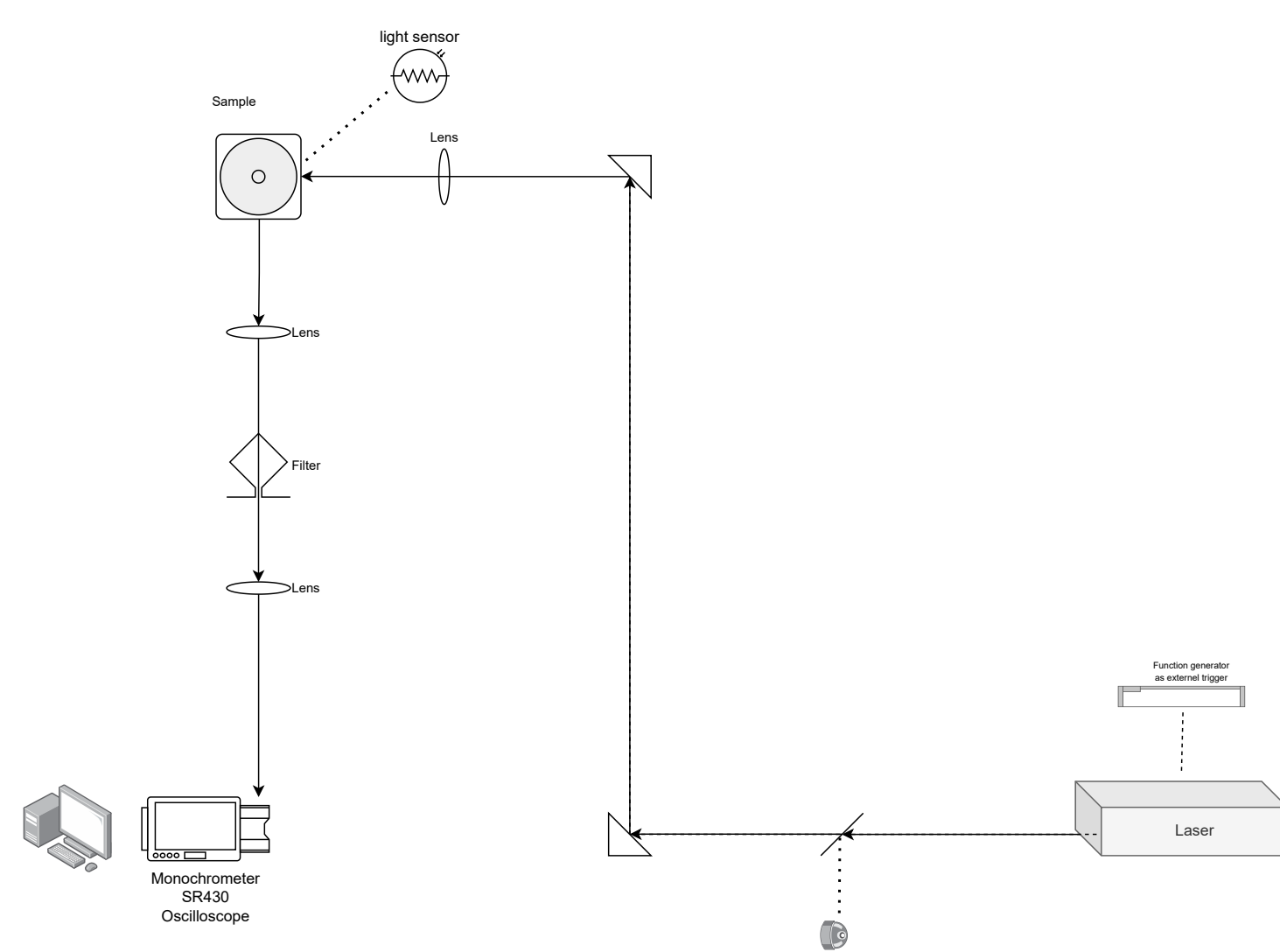
UV light

3. DFT Calculated 4f electronic levels



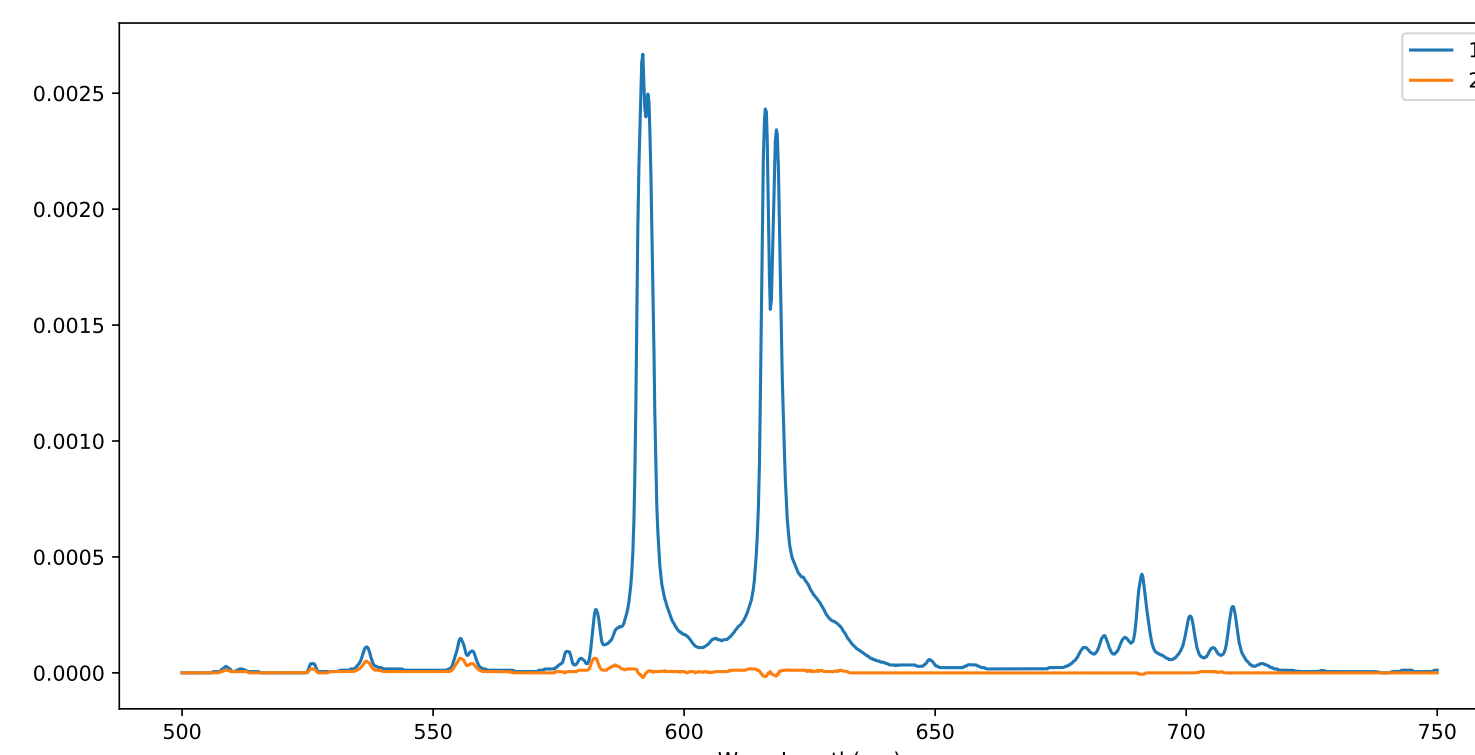
3. Experimental Setup

Measured the fluorescence of the Eu at 618 nm after pulsed excitation, with 337 nm and 266 nm.

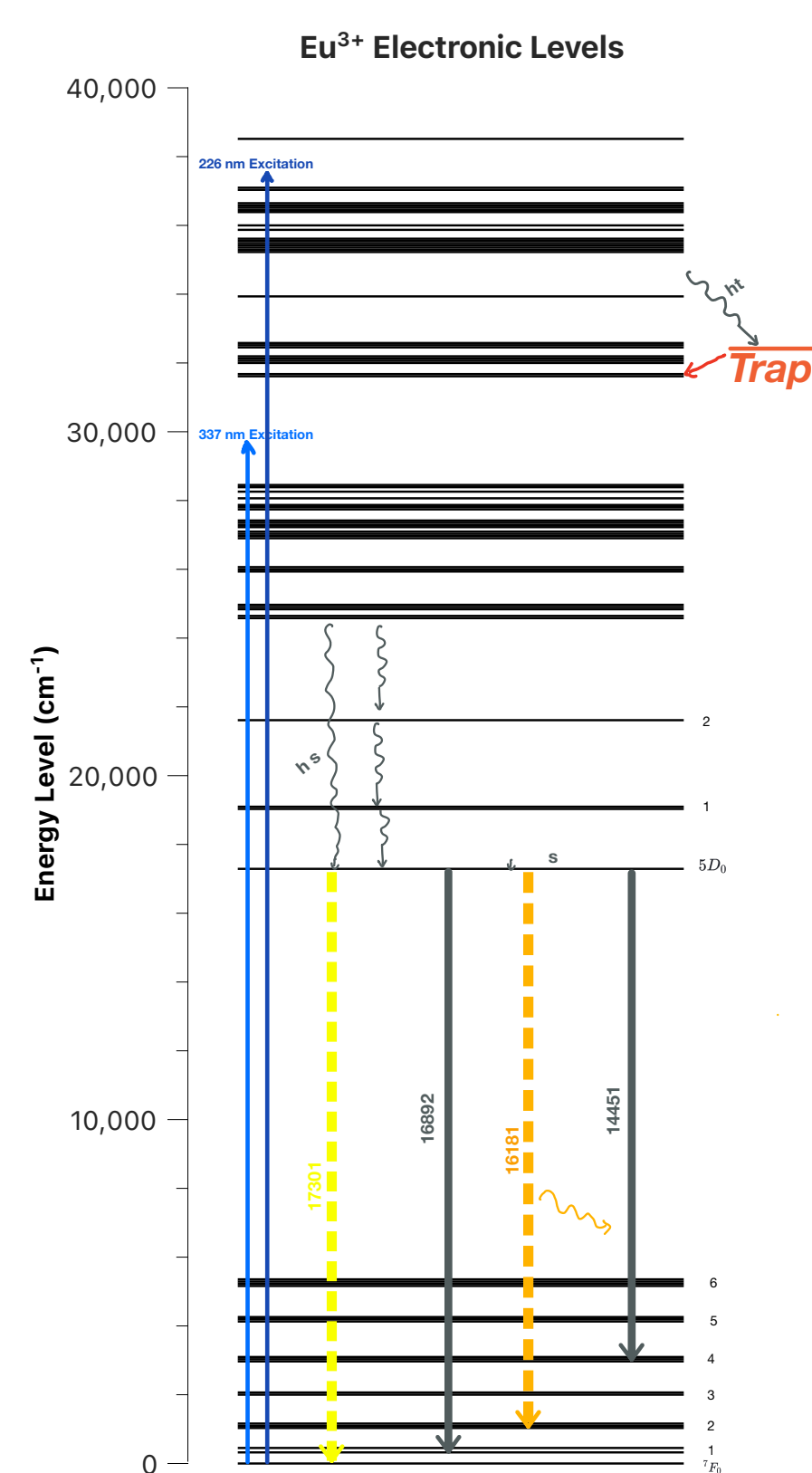


4. Measurement Results

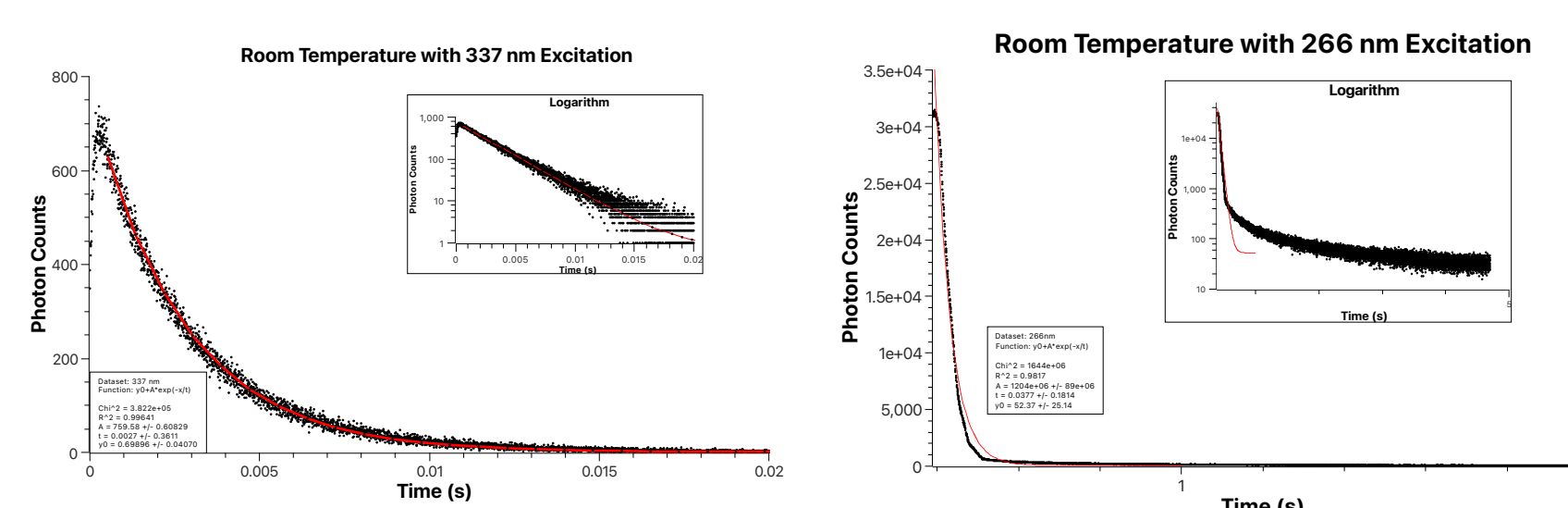
1. Emission (266 nm excitation)



2. Electronic Levels, transitions with charge trap model.



3. Typical lifetime measurement results

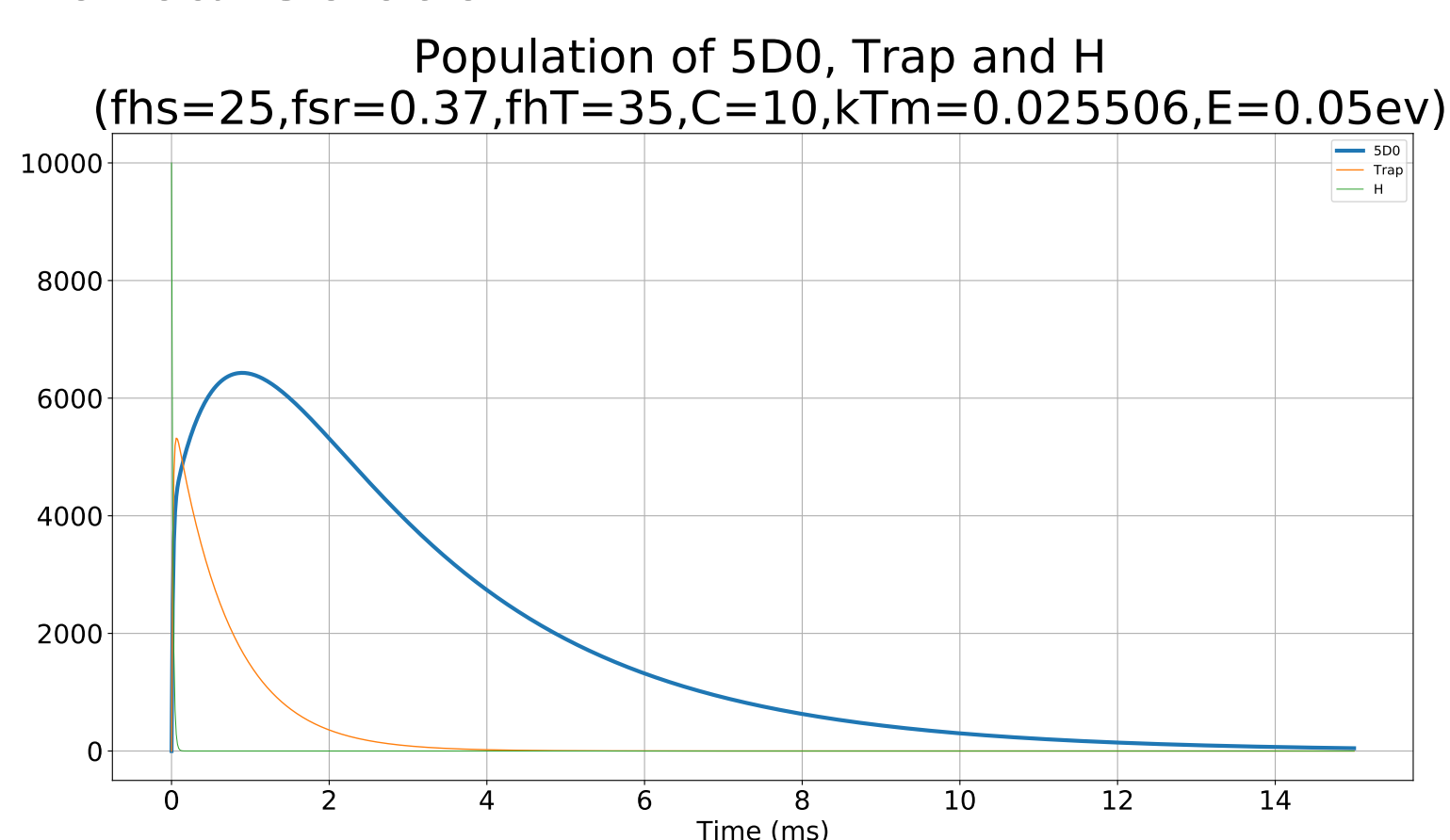


5. Proposed Model

$$\begin{aligned}\dot{S} &= Hf_{hs} + Tf_{ts} - Sf_{sr} \\ \dot{T} &= Hf_{ht} - Tf_{ts} \\ \dot{H} &= -H(f_{ht} + f_{hs})\end{aligned}\quad (1)$$

S is the population of Eu in the 5D_0 excited state. T is the population of the traps. H is the population of any higher energy state. f is the rate of transmission. (Population: the number of Eu in each state)

• Numerical Solution



Blue is the S and Orange is the Trap population

• Analytical Solution

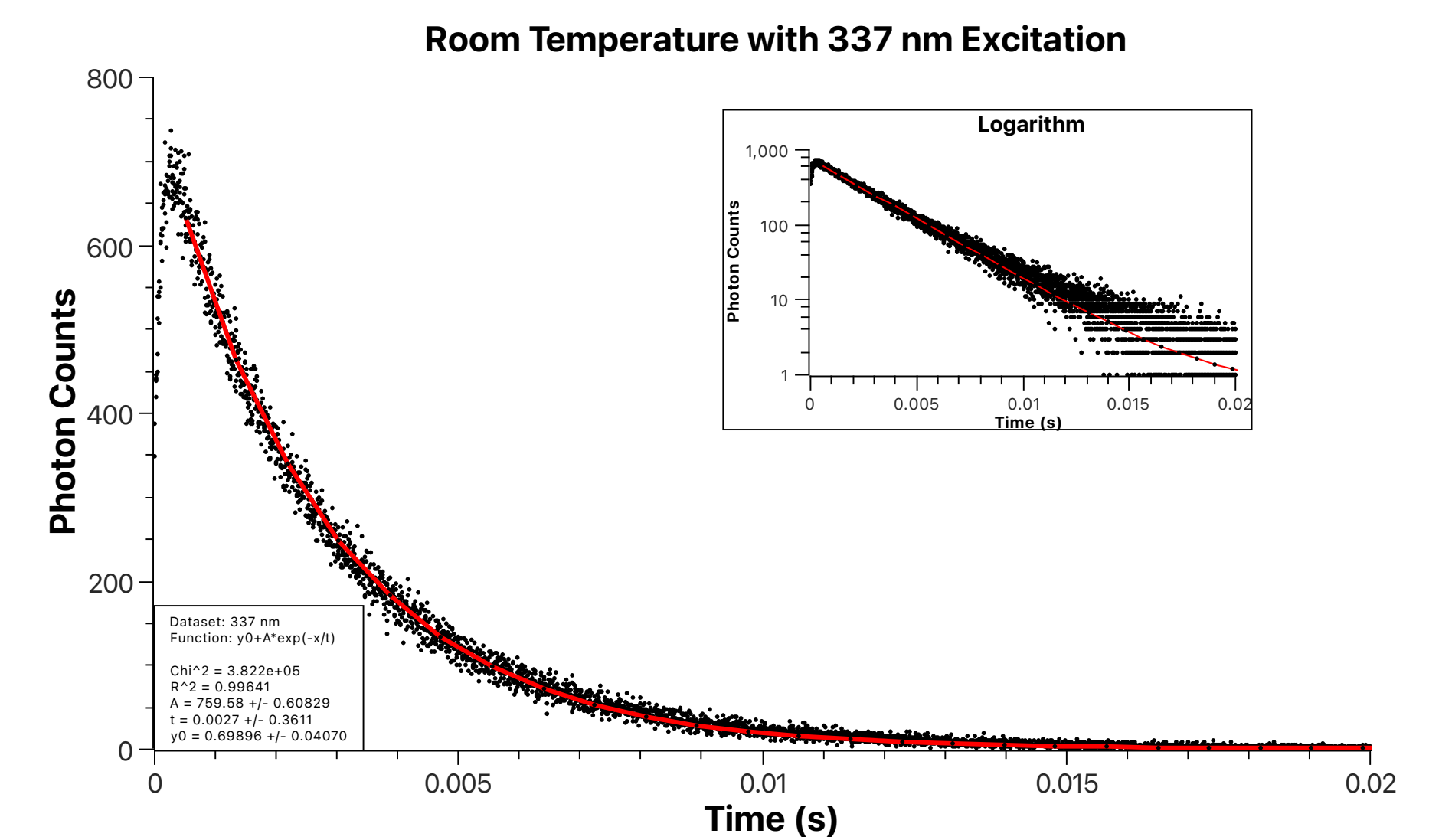
Ignore initial filling of S and T from H

$$S(t) = C_s e^{-f_{sr}t} + \left(\frac{f_{ts}}{f_{sr} - f_{ts}} \right) C_T e^{-f_{ts}t} \quad (2)$$

$$\frac{dS}{dt} = -k_1 e^{-f_{sr}t} - k_2 e^{-f_{ts}t} \quad (3)$$

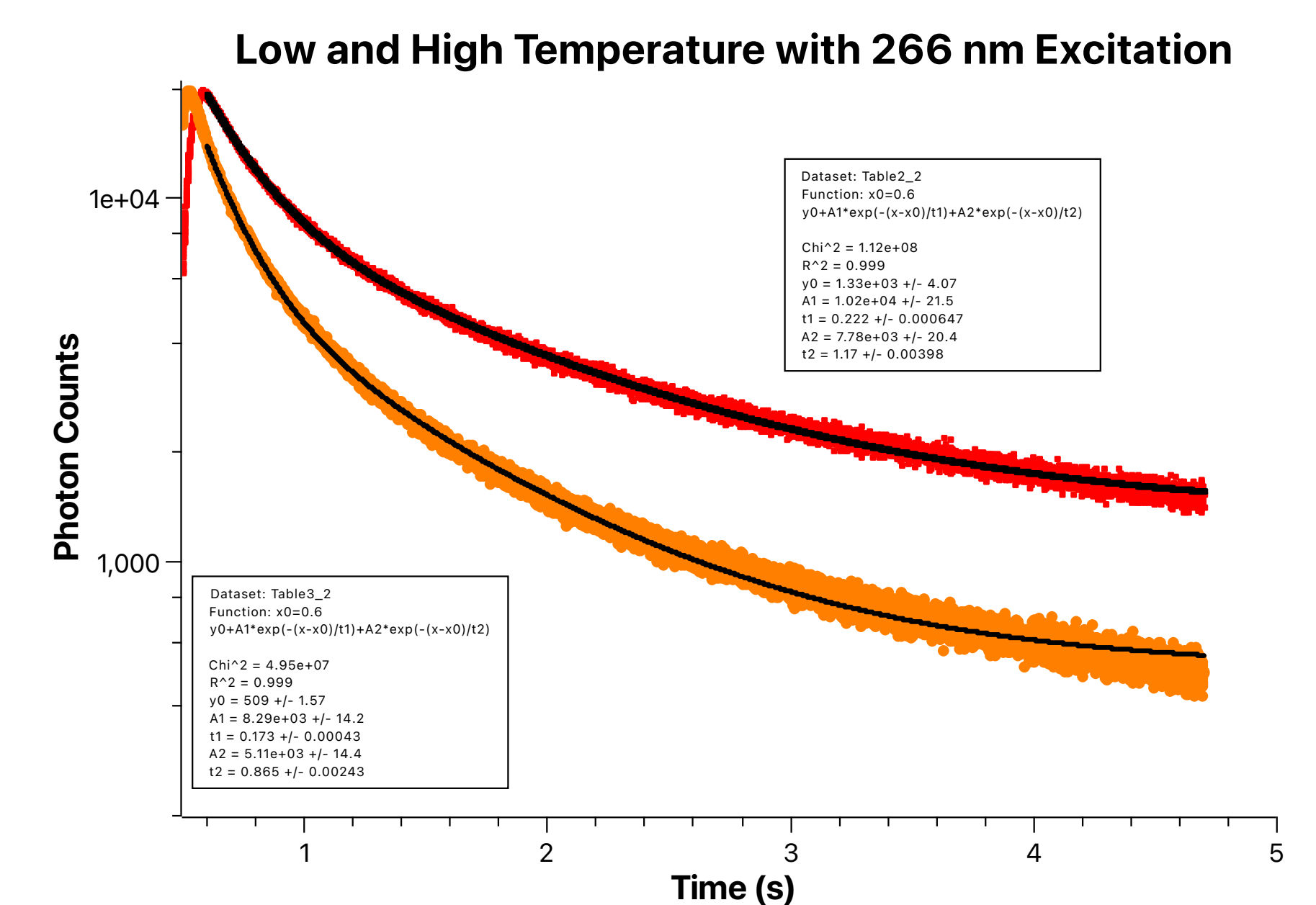
6. Model Fit with 337 nm

Asuming $k_2 \ll k_1$ and $f_{ts} \gg 1$



- The 337 nm excitation does not contribute to phosphorescence.
- Radiative lifetime of Europium is 2.7 ms.
- The rise of population after pulse is observed.

7. Model Fit with 226 nm, 78K, 250K...



- The biexponential model fits.
- The phosphorescence strongly depends on temperature, with 35% increase in the T to S transfer rate (f_{ts}) over the 78-255K range.

8. Conclusions

- Sample of LaAlO₃:Eu³⁺ have been made by combustion synthesis.
- The emission spectrum was used to identify the transition between 4f levels and compared to DFT calculation with good agreement
- The decay of both phosphorescence and fluorescence emission has been measured.
- Both emissions are well-described by simple biexponential model.
- The trap to impurity transfer rate has been found to decrease at lower temperatures.

References

- [1] M. Lastusaari P. Gluchowski, W. Strek and J. Holsa. Optically stimulated persistent luminescence of europium-doped LaAlO₃ nanocrystals. *Phys.Chem.Chem.Phys.*, 17:17246–17252, 2015.
- [2] C. Furetta and G. Kitis. Review Models in thermoluminescence. *Journal of Materials Science*, 39:2277–2294, 2004.



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